# ABBREVIATIONS

| | | | |
|---|---|---|---|
| A/C | ABOVE CEILING | HWRR | HOT WATER REVERSE RETURN |
| AAV | AIR ADMITTANCE VALVE | HWS | HOT WATER SUPPLY |
| AD | ACCESS DOOR | | |
| ADJ | ADJUSTABLE | ID | INSIDE DIMENSION |
| AFF | ABOVE FINISHED FLOOR | IE | INVERTED ELEVATION |
| AUTO | AUTOMATIC | IN | INCHES |
| B/G | BELOW GRADE | KW | KILOWATTS |
| B/S | BELOW SLAB | | |
| BAL | BALANCING | LB | POUNDS |
| BCO | BASE CLEANOUT | LP | LIQUID PROPANE GAS |
| BHP | BRAKE HORSEPOWER | LWR | LOOP WATER RETURN |
| BOD | BASIS OF DESIGN | LWS | LOOP WATER SUPPLY |
| CO | CLEANOUT | MAX | MAXIMUM |
| COND | CONDENSATE | MIN | MINIMUM |
| CW | COLD WATER (DOMESTIC) | MFR | MANUFACTURER |
| DN | DOWN | NC | NORMALLY CLOSED |
| DR | DRAIN | NFGH | NON-FREEZE GROUND HYDRANT |
| DWG | DRAWING | NFWH | NON-FREEZE WALL HYDRANT |
| | | NG | NATURAL GAS |
| EA | EACH | NO | NORMALLY OPEN |
| ECC | ECCENTRIC | NOM | NOMINAL |
| EFF | EFFICIENCY | | |
| EOD | EMERGENCY OVERFLOW DRAIN | OD | OUTSIDE DIMENSION |
| | | OD | OVERFLOW DRAIN |
| EWT | ENTERING WATER TEMP. | | |
| EX | EXISTING | PSI | POUNDS PER SQUARE INCH |
| F | FAHRENHEIT | RAD | RADIUS |
| FCO | FLOOR CLEANOUT | RED | REDUCER |
| FD | FLOOR DRAIN | | |
| FLR | FLOOR | SAN | SANITARY |
| FOB | FLAT ON BOTTOM | SQ | SQUARE |
| FOT | FLAT ON TOP | ST | STORM |
| FPM | FEET PER MINUTE | | |
| FPS | FEET PER SECOND | TEMP | TEMPERATURE |
| FT | FEET | TYP | TYPICAL |
| G | GATE | UON | UNLESS OTHERWISE NOTED |
| GA | GAUGE | | |
| GCO | GRADE CLEANOUT | V | VENT |
| GL | GLOBE | VA | VALVE |
| GPM | GALLONS PER MINUTE | VTR | VENT THRU ROOF |
| GW | GREASE WASTE | | |
| | | WC | WATER COLUMN |
| HD | HUB DRAIN | WCO | WALL CLEANOUT |
| HP | HORSEPOWER | WHA | WATER HAMMER ARRESTOR |
| HW | HOT WATER (DOMESTIC) | WT | WEIGHT |
| HWR | HOT WATER RETURN | W | WASTE |

# APPLICABLE CODES

- International Building Code, 2024 Edition, with Georgia Amendments (2026)
- International Fire Code, 2024 Edition
- International Plumbing Code, 2024 Edition, with Georgia Amendments (2026)
- International Fuel Gas Code, 2024 Edition, with Georgia Amendments (2026)
- International Energy Conservation Code, 2015 Edition, with Georgia Supplements and Amendments

# PLUMBING SHEET INDEX:


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# SPECIFICATIONS

**PART 1 - GENERAL**

1. SUBMITTALS
   A. PRODUCT DATA: FOR EACH TYPE OF PRODUCT INDICATED.

**PART 2 - PRODUCTS**

1. PIPING MATERIALS
   A. COMPLY WITH REQUIREMENTS IN "PLUMBING PIPING SCHEDULE" ARTICLE FOR APPLICATIONS OF PIPE, TUBE, AND FITTING MATERIALS, AND FOR JOINING METHODS, INSULATION, AND HANGING REQUIREMENTS FOR SPECIFIC SERVICES, SERVICE LOCATIONS, AND PIPE SIZES.
2. VALVES
   A. NSF COMPLIANCE: NSF 61 FOR VALVE MATERIALS FOR POTABLE -WATER SERVICE.
   B. WATER SHUTOFF SERVICE: TWO-PIECE, FULL PORT, BRASS BALL VALVES WITH BRASS TRIM.
   C. CHECK VALVES: BRASS SILENT CHECK VALVES WITH NONMETALLIC DISC SUITABLE FOR HORIZONTAL OR VERTICAL INSTALLATION.
   D. FUEL GAS SHUTOFF SERVICE: TWO-PIECE, FULL PORT, BRASS BALL VALVES WITH BRASS TRIM, UL LISTED FOR GAS SERVICE.
   E. FUEL GAS PRESSURE REGULATORS: COMMERCIAL STYLE REGULATORS WITH INTERNAL PRESSURE RELIEF VALVES.
3. HANGERS AND SUPPORTS
   A. PROVIDE HANGERS INDICATED ON DRAWINGS.
   B. WALL SUPPORTS
      - PIPING 3" AND SMALLER: STEEL RISER CLAMP.
   C. VERTICAL SUPPORT: STEEL RISER CLAMP.

**PART 3 - EXECUTION**

1. PIPING INSTALLATION
   A. INSTALL ESCUTCHEONS AT EACH WALL, FLOOR, AND CEILING PENETRATION IN EXPOSED FINISHED LOCATIONS AND WITHIN CABINETS AND MILLWORK. USE DEEP PATTERN ESCUTCHEONS WHERE REQUIRED TO CONCEAL PROTRUDING PIPE FITTINGS.
   B. INSTALL SLEEVES FOR PIPING PENETRATIONS OF WALLS, CEILINGS, AND FLOORS.
   C. INSTALL SLEEVE SEALS FOR PIPING PENETRATIONS OF CONCRETE WALLS AND SLABS.
   D. MAKE SOLDERED JOINTS USING LEAD FREE SOLDER AND A NON-CORROSIVE, PASTE-TYPE FLUX. CORE SOLDER IS NOT PERMITTED SOLDER SHALL BE SOLID STRING OR WIRE TYPE. WHERE SOLDERED COPPER PIPING IS CONNECTED TO THREADED BRASS PIPING, USE CAST BRASS ADAPTOR.
   E. PLACE PLUGS IN ENDS OF UNCOMPLETED PIPING AT END OF EACH DAY OR WHEN WORK STOPS.
   F. MINIMUM SLOPE OF HORIZONTAL DRAIN PIPING SHALL BE 1/4" PER FOOT FOR PIPING 2-1/2" OR LESS, 1/8" PER FOOT FOR PIPING 3" TO 6", AND 1/16" PER FOOT FOR PIPING 8" OR LARGER.
   G. MAKE CHANGES IN DIRECTION IN DRAIN PIPING WITH INDIVIDUAL EIGHT BENDS AND WYES.
   H. QUARTER BENDS, SWEEPS, TEES, AND COMBINATION WYE & EIGHT BEND FITTINGS ARE NOT PERMITTED FOR USE ON DRAIN PIPING.
   I. EXTEND RIGID GAS PIPING TO EXTERIOR GAS APPLIANCES AND INSTALL SHUTOFF VALVE, DIRT LEG, AND UNION AT EACH APPLIANCE.
2. JOINT CONSTRUCTION
   A. REAM ENDS OF PIPES AND TUBES AND REMOVE BURRS. BEVEL PLAIN ENDS OF STEEL PIPE.
   B. REMOVE SCALE, SLAG, DIRT, AND DEBRIS FROM INSIDE AND OUTSIDE OF PIPES, TUBES, AND FITTINGS BEFORE ASSEMBLY.
   C. THREADED JOINTS: THREAD PIPE WITH TAPERED PIPE THREADS ACCORDING TO ASME B1.20.1. CUT THREADS FULL AND CLEAN USING SHARP DIES. REAM THREADED PIPE ENDS TO REMOVE BURRS AND RESTORE FULL ID. JOINT PIPE FITTINGS AND VALVES AS FOLLOWS:
      - 1. APPLY APPROPRIATE TAPE OR THREAD COMPOUND TO EXTERNAL PIPE THREADS.
      - DAMAGED THREADS: DO NOT USE PIPE OR FITTINGS WITH THREADS THAT ARE CORRODED OR DAMAGED.
   D. DISSIMILAR-MATERIAL PIPING JOINTS: MAKE JOINTS USING ADAPTERS COMPATIBLE WITH MATERIALS OF BOTH PIPING SYSTEMS.
3. HANGERS AND SUPPORT INSTALLATION
   A. SUPPORT PIPES THROUGHOUT BUILDING, BOTH HORIZONTALLY AND VERTICAL IN ACCORDANCE WITH REQUIREMENTS HEREIN AND AS SHOWN ON THE DRAWINGS. DO NOT USE FASTENERS WHICH PENETRATE THE ROOF DECK.
   B. PROVIDE COPPER PLATED HANGERS AND SUPPORTS FOR UNINSULATED COPPER PIPING. PROVIDE PLASTIC INSERTS FOR UNINSULATED COPPER PIPING PENETRATING METAL STUDS.
   C. IN AREAS WITHOUT CEILINGS, SECURE INSULATION SHIELDS TO INSULATION WITH PRESSURE SENSITIVE TAPE AT EACH END OF SHIELD. INSTALL HANGERS TO PROVIDE MINIMUM ½" CLEAR SPACE BETWEEN FINISHED COVERING AND ADJACENT WORK.
   D. PLACE A HANGER WITHIN ONE FOOT OF EACH HORIZONTAL ELBOW.
   E. SUPPORT VERTICAL PIPE RUNS AT ROOF, AT FLOOR, AND AT MAXIMUM 15-FOOT INTERVALS.
   F. SPACE SUPPORTS NOT MORE THAN FIVE FEET APART AT VALVES, STRAINERS OR PIPING ACCESSORIES LARGER THAN 2".
   G. INSTALL LATERAL BRACING WITH PIPE HANGERS AND SUPPORTS TO PREVENT SWAYING.
   H. INSTALL HANGERS AND SUPPORTS TO ALLOW CONTROLLED THERMAL AND SEISMIC MOVEMENT OF PIPING SYSTEMS, TO PERMIT FREEDOM OF MOVEMENT BETWEEN PIPE ANCHORS, AND TO FACILITATE ACTION OF EXPANSION JOINTS, EXPANSION LOOPS, EXPANSION BENDS, AND SIMILAR UNITS.
   I. ADJUSTING
      - HANGER ADJUSTMENTS: ADJUST HANGERS TO DISTRIBUTE LOADS EQUALLY ON ATTACHMENTS AND TO ACHIEVE INDICATED SLOPE OF PIPE.
      - LOAD DISTRIBUTION: ADJUST HANGERS AND SUPPORTS SO THAT PIPING LIVE AND DEAD LOADS AND STRESSES FROM MOVEMENT WILL NOT BE TRANSMITTED TO CONNECTED EQUIPMENT.
      - PIPE SLOPES: ADJUST HANGERS AND SUPPORTS TO PROVIDE INDICATED PIPE SLOPES AND TO NOT EXCEED MAXIMUM PIPE DEFLECTIONS ALLOWED BY ASME B31.9 FOR BUILDING SERVICES PIPING.
      - TRIM EXCESS LENGTH OF CONTINUOUS - THREAD HANGER AND SUPPORT RODS TO 1-1/2 ".
4. PIPE INSULATION INSTALLATION
   A. INSULATE FITTINGS AND VALVES WITH PREFORMED INSULATION FITTINGS WITH PVC JACKET.
5. VALVE AND SPECIALTIES INSTALLATION
   A. PROVIDE DEEP SEAL P-TRAPS FOR FLOOR DRAINS AND MOP BASINS.
   B. FOR CLEANOUTS LOCATED IN CONCEALED PIPING, INSTALL CLEANOUT WALL ACCESS COVERS OF TYPES INDICATED, WITH FRAME AND COVER FLUSH WITH FINISHED WALL.
   C. PROTECT DRAINS DURING CONSTRUCTION PERIOD TO AVOID CLOGGING WITH DIRT OR DEBRIS AND TO PREVENT DAMAGE FROM TRAFFIC OR CONSTRUCTION WORK.

# GENERAL NOTES:

A. PROVIDE ALL MATERIALS AND EQUIPMENT AND PERFORM ALL LABOR REQUIRED TO INSTALL COMPLETE AND OPERABLE PLUMBING SYSTEMS AS INDICATED ON THE DRAWINGS, AS SPECIFIED AND AS REQUIRED BY CODE.

B. CONTRACT DOCUMENT DRAWINGS FOR PLUMBING WORK ARE DIAGRAMMATIC AND ARE INTENDED TO CONVEY SCOPE AND GENERAL ARRANGEMENT ONLY.

C. INSTALL ALL PLUMBING EQUIPMENT AND APPURTENANCES IN ACCORDANCE WITH MANUFACTURERS' RECOMMENDATIONS, CONTRACT DOCUMENTS, AND APPLICABLE CODES AND REGULATIONS.

D. PROVIDE VIBRATION ISOLATION FOR ALL PLUMBING EQUIPMENT TO PREVENT TRANSMISSION OF VIBRATION TO BUILDING STRUCTURE.

E. PROVIDE VIBRATION ISOLATORS FOR ALL PIPING SUPPORTS CONNECTED TO AND WITHIN 50 FEET OF ISOLATED EQUIPMENT (EXCEPT AT BASE ELBOW SUPPORTS AND ANCHOR POINTS) THROUGHOUT MECHANICAL EQUIPMENT ROOMS. DO THE SAME FOR SUPPORTS OF STEAM MAINS WITHIN 50 FEET OF BOILER OR PRESSURE REDUCING VALVES.

F. PROVIDE VIBRATION ISOLATORS FOR ALL PIPING SUPPORTS OF STEAM MAINS WITHIN 50 FEET OF BOILERS AND PRESSURE REDUCING VALVES.

G. THE LOCATION OF EXISTING UNDERGROUND UTILITIES IS SHOWN IN AN APPROXIMATE WAY ONLY. THE CONTRACTOR SHALL DETERMINE THE EXACT LOCATION OF ALL EXISTING UTILITIES BEFORE COMMENCING WORK. THE CONTRACTOR SHALL PAY FOR AND REPAIR ALL DAMAGES CAUSED BY FAILURE TO EXACTLY LOCATE AND PRESERVE ANY AND ALL UNDERGROUND UTILITIES UNLESS OTHERWISE INDICATED.

H. COORDINATE CONSTRUCTION OF ALL PLUMBING WORK WITH ARCHITECTURAL, STRUCTURAL, CIVIL, ELECTRICAL WORK, ETC., SHOWN ON OTHER CONTRACT DOCUMENT DRAWINGS.

I. ALL TESTS SHALL BE COMPLETED BEFORE ANY PLUMBING EQUIPMENT OR PIPING INSULATION IS APPLIED.

J. LOCATE ALL TEMPERATURE, PRESSURE, AND FLOW MEASURING DEVICES IN ACCESSIBLE LOCATIONS WITH STRAIGHT SECTION OF PIPE AND DOWNSTREAM AS RECOMMENDED BY THE MANUFACTURER FOR GOOD ACCURACY.

K. WHERE TWO OR MORE ITEMS OF THE SAME TYPE OF EQUIPMENT ARE REQUIRED, THE PRODUCT OF ONE MANUFACTURER SHALL BE USED.

L. REINFORCEMENT, DETAILING, AND PLACEMENT OF CONCRETE SHALL CONFORM TO ASTM 315 AND ACI 318. CONCRETE SHALL CONFORM TO ASTM C94. CONCRETE WORK SHALL CONFORM TO ACI 318, PART ENTITLED "CONSTRUCTION REQUIREMENTS." COMPRESSIVE STRENGTH IN 28 DAYS SHALL BE 3,000 PSI. TOTAL AIR CONTENT OF EXTERIOR CONCRETE SHALL BE BETWEEN 5 AND 7 PERCENT BY VOLUME. SLUMP SHALL BE BETWEEN 3 AND 4 INCHES. CONCRETE SHALL BE CURED FOR 7 DAYS AFTER PLACEMENT.

M. COORDINATE ALL EQUIPMENT CONNECTIONS WITH MANUFACTURERS' CERTIFIED DRAWINGS. COORDINATE AND PROVIDE ALL PIPING TRANSITIONS REQUIRED FOR FINAL EQUIPMENT CONNECTIONS TO FURNISHED EQUIPMENT. FIELD VERIFY AND COORDINATE ALL PIPING DIMENSIONS BEFORE FABRICATION.

N. ALL CONTROL WIRE AND CONDUIT SHALL COMPLY WITH THE NATIONAL ELECTRIC CODE AND DIVISION 16 OF THE SPECIFICATION.

O. WHERE BEAMS ARE INDICATED TO BE PENETRATED WITH PIPING, COORDINATE PIPING LAYOUT WITH BEAM OPENING SIZE AND OPENING LOCATIONS. COORDINATION SHALL BE DONE PRIOR TO CUTTING OF PIPING, OR FABRICATION OF BEAMS.

P. THE LOCATIONS OF ALL ITEMS SHOWN ON THE DRAWINGS OR CALLED FOR IN THE SPECIFICATIONS THAT ARE NOT DEFINITELY FIXED BY DIMENSIONS ARE APPROXIMATE ONLY. THE EXACT LOCATIONS NECESSARY TO SECURE THE BEST CONDITIONS AND RESULTS MUST BE DETERMINED BY THE PROJECT SITE CONDITIONS AND SHALL HAVE THE APPROVAL OF THE ENGINEER BEFORE BEING INSTALLED. DO NOT SCALE DRAWINGS.

Q. PROVIDE ACCESS PANELS FOR INSTALLATION IN WALLS AND CEILINGS, WHERE REQUIRED, TO SERVICE CONCEALED PLUMBING EQUIPMENT OR VALVES. ACCESS PANELS SHALL BE TURNED OVER TO GENERAL CONTRACTOR FOR INSTALLATION.

R. ALL EQUIPMENT, PIPING, ETC., SHALL BE SUPPORTED AS DETAILED, SPECIFIED, AND REQUIRED TO PROVIDE A VIBRATION FREE INSTALLATION.

S. ALL PIPING AND EQUIPMENT SUPPORTED FROM STRUCTURAL STEEL SHALL BE COORDINATED WITH GENERAL CONTRACTOR. ALL ATTACHMENTS TO STEEL BAR JOISTS, TRUSSES, OR JOIST GIRDERS SHALL BE AT PANEL POINTS. PROVIDE BEAM CLAMPS MEETING MSS STANDARDS. WELDING TO STRUCTURAL MEMBERS SHALL NOT BE PERMITTED. THE USE OF C-CLAMPS SHALL NOT BE PERMITTED.

T. PLUMBING EQUIPMENT AND PIPING SHALL NOT BE SUPPORTED FROM METAL DECK.

U. LOCATIONS AND SIZES OF ALL FLOOR, WALL, AND ROOF OPENINGS SHALL BE COORDINATED WITH ALL OTHER TRADES INVOLVED. ROOF PENETRATIONS AND FLASHING SHALL BE DONE BY THE ROOFING CONTRACTOR.

V. ALL OPENINGS IN FIRE WALLS DUE TO PIPING, CONDUIT, ETC., SHALL BE FIRE STOPPED WITH A PRODUCT SIMILAR TO 3M OR APPROVED EQUAL. REFER TO TYPICAL DETAILS FOR PIPING AND EQUIPMENT INSTALLATION.

# PLUMBING NOTES:

A. PROVIDE ALL MATERIALS AND EQUIPMENT AND PERFORM ALL LABOR REQUIRED TO INSTALL COMPLETE AND OPERABLE PLUMBING SYSTEMS AS INDICATED ON THE DRAWINGS, AS SPECIFIED AND AS REQUIRED BY CODE.

B. RUN ALL SOIL WASTE AND VENT PIPING WITH 2% MINIMUM GRADE UNLESS OTHERWISE NOTED. HORIZONTAL VENT PIPING SHALL BE GRADED TO DRIP BACK TO THE SOIL OR WASTE PIPE BY GRAVITY.

C. ELEVATIONS AS SHOWN ON THE DRAWINGS ARE TO THE CENTERLINE OF ALL PRESSURE PIPING AND TO THE INVERT OF ALL GRAVITY PIPING.

D. ADJUST SEWER INVERTS TO KEEP TOPS OF PIPE IN LINE WHERE PIPE SIZE CHANGES.

E. MAINTAIN A MINIMUM OF 3'6" OF GROUND COVER OVER ALL UNDERGROUND WATER MAINS AND A MINIMUM OF 3'0" OF GROUND COVER OVER ALL UNDERGROUND SEWERS AND DRAINS.

F. PROVIDE SHUTOFF VALVES IN ALL DOMESTIC WATER PIPING SYSTEM BRANCHES IN WHICH BRANCH PIPING SERVES TWO OR MORE FIXTURES.

G. UNLESS OTHERWISE NOTED, ALL DOMESTIC COLD AND HOT WATER PIPING SHALL BE 1⁄2 INCH SIZE.

H. UNLESS OTHERWISE NOTED, ALL PIPING IS OVERHEAD, TIGHT TO UNDERSIDE OF SLAB, WITH SPACE FOR INSULATION IF REQUIRED.

I. INSTALL PIPING SO THAT ALL VALVES, STRAINERS, UNIONS, TRAPS, FLANGES, AND OTHER APPURTENANCES REQUIRING ACCESS ARE ACCESSIBLE.

J. INSTALL PIPING VALVES FEEDING MULTIPLE FIXTURES AT CONVENIENT CENTRAL LOCATION AND APPROXIMATELY 6"-12" ABOVE FINISHED HARD GYPSUM BOARD CEILINGS FOR EASE OF ACCESS THROUGH SMALLER CEILING ACCESS PANELS.

K. WHERE DOMESTIC COLD AND HOT WATER PIPING DROPS INTO A PIPE CHASE, THE SIZE SHOWN FOR THE PIPE DROPS SHALL BE USED TO THE LAST FIXTURE.

L. INSTALL ALL PIPING WITHOUT FORCING OR SPRINGING.

M. ALL PIPING SHALL CLEAR DOORS AND WINDOWS.

N. ALL PIPING SHALL GRADE TO LOW POINTS. PROVIDE HOSE END DRAIN VALVES AT THE BOTTOM OF ALL RISERS AND LOW POINTS.

O. UNIONS AND/OR FLANGES SHALL BE INSTALLED AT EACH PIECE OF EQUIPMENT, IN BYPASSES, AND IN LONG PIPING RUNS (100 FEET OR MORE) TO PERMIT DISASSEMBLY FOR ALTERATION AND REPAIRS.

P. ALL VALVES SHALL BE ADJUSTED FOR SMOOTH AND EASY OPERATION.

Q. ALL VALVES (EXCEPT CONTROL VALVES) AND STRAINERS SHALL BE FULL SIZE OF PIPE BEFORE REDUCING SIZE TO MAKE CONNECTIONS TO EQUIPMENT AND CONTROLS.

R. PROVIDE CHAINWHEEL OPERATORS FOR ALL VALVES IN EQUIPMENT ROOMS MOUNTED GREATER THAN 7'-0 " ABOVE FLOOR LEVEL; CHAIN SHALL EXTEND TO 7'-0" ABOVE FLOOR LEVEL.

S. PROVIDE ALL PLUMBING FIXTURES AND EQUIPMENT WITH ACCESSIBLE STOPS.

T. UNLESS OTHERWISE NOTED, DRAINS SHALL BE INSTALLED AT THE LOW POINT OF ROOFS, AREAWAYS, FLOORS, ETC.

U. PROVIDE CLEANOUTS IN SANITARY AND STORM DRAINAGE SYSTEMS AT ENDS OF RUNS, AT CHANGES IN DIRECTION, NEAR THE BASE OF STACKS, EVERY 50 FEET IN HORIZONTAL RUNS AND ELSEWHERE AS INDICATED.

V. ALL CLEANOUTS SHALL BE FULL SIZE OF PIPE FOR PIPE SIZES 6 INCHES AND SMALLER AND SHALL BE 6 INCHES FOR PIPE SIZES LARGER THAN 6 INCHES.

W. ALL BALANCING VALVES AND BUTTERFLY VALVES SHALL BE PROVIDED WITH POSITION INDICATORS AND MAXIMUM ADJUSTABLE STOPS (MEMORY STOPS).

X. ALL VALVES SHALL BE INSTALLED SO THAT VALVE REMAINS IN SERVICE WHEN EQUIPMENT OR PIPING ON EQUIPMENT SIDE OF VALVE IS REMOVED.

Y. ALL PIPING WORK SHALL BE COORDINATED WITH ALL TRADES INVOLVED. OFFSETS IN PIPING AROUND OBSTRUCTIONS SHALL BE PROVIDED AT NO ADDITIONAL COST TO THE OWNER.

Z. PROVIDE FLEXIBLE CONNECTIONS IN ALL PIPING SYSTEMS CONNECTED TO PUMPS AND OTHER EQUIPMENT WHICH REQUIRE VIBRATION ISOLATION. FLEXIBLE CONNECTIONS SHALL BE PROVIDED AS CLOSE TO THE EQUIPMENT AS POSSIBLE OR AS INDICATED ON THE DRAWINGS.

AA. PLUMBING CONTRACTOR TO ENSURE THAT VENT STACK ROUTING FOR WALL MOUNTED LAVATORY FIXTURES DO NOT CONFLICT WITH SINK CARRIER INSTALLATION.

**DISCLAIMER NOTE**: NOT ALL SPECIFICATIONS OR NOTES ON THIS SHEET MAY APPLY TO THIS SPECIFIC PROJECT. REFER TO PROJECT MECHANICAL PLANS FOR SCOPE OF WORK APPLICABLE TO THIS PROJECT.


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| Rev | Date | Comments |
|---|---|---|
| | 07/02/2026 | PERMIT SET |

Client:

**FIFTH THIRD BANK**

Project Number: 26032
Project Name:

**FIFTH THIRD BANK WATSON 1,900 SF**

2711 WATSON BOULEVARD
WARNER ROBBINS, GA 31093

Key Plan:

Sheet Title:

**PLUMBING SPECIFICATIONS AND NOTES**

Sheet Number:

**P-001**

6/30/2026 4:55:07 PM

# PLUMBING PIPING SCHEDULE:

| SYSTEM | LOCATION | SIZE | PIPE MATERIAL | JOINTS | FITTINGS | HORIZONTAL HANGER SPACING (FT.) | VERTICAL SPACING (FT.) | NOTES |
|---|---|---|---|---|---|---|---|---|
| DOMESTIC COLD AND HOT WATER | ABOVE GRADE | 1-1/4" AND SMALLER | TYPE L HARD COPPER | SOLDER / PROPRESS | WROUGHT COPPER | 6 | 10 | - |
| | | 1-1/2" AND LARGER | | | | 10 | 10 | - |
| SANITARY SEWER | ALL | 1-1/4" AND SMALLER | HUBLESS CAST IRON | HUBLESS | HUBLESS CAST IRON | 5 | 15 | 1, 2 |
| | | 1-1/2" AND LARGER | DWV HARD COPPER | SOLDER | CAST BRONZE OR WROUGHT COPPER | 10 | 10 | - |
| | | 1-1/4" | | | | 6 | 10 | - |
| | | ALL | PVC SHED 40 DWV | SOLVENT WELD | DWV SOCKET | 4 | 10 | 3, 4 |
| NATURAL GAS | ABOVE GRADE | 1/2" | SCHD 40 BLACK IRON | THREADED, WELDED, MEGAPRESS | MALLEABLE IRON, MEGAPRESS | 6 | 15 | - |
| | | 3/4" THRU 1-1/2" | | | | 8 | 15 | - |
| | | 2" AND LARGER | | WELDED, MEGAPRESS | BUTT WELDED, MEGAPRESS | 10 | 15 | - |

**GENERAL NOTES:**

A. ALL PRODUCTS USED IN DOMESTIC COLD AND HOT WATER PIPING SYSTEMS MUST BE NSF/ANSI 61 CERTIFIED AND BE LEAD FREE. LEAD FREE PRODUCTS MUST BE LISTED AND LABELED BY A THIRD PARTY AS LEAD FREE SUCH THAT THE WETTED SURFACE MATERIALS HAVE NO MORE THAN 0.25% LEAD CONTENT.

B. INSULATION TO BE FIBERGLASS, POLYMER FOAM, OR ELASTOMERIC PREFORMED PIPE INSULATION WITH SELF SEALING ADHESIVE LAP.

C. ANY PIPING TO BE RECEIVING FIBERGLASS INSULATION MUST ALSO HAVE VAPOR RETARDED JACKET, SEAL JOINTS WITH 2" WIDE PRESSURE SENSITIVE VAPOR RETARDED TAPE.

D. INSULATION MUST BE INSTALLED ACCORDING TO MANUFACTURER'S REQUIREMENTS.

E. PIPE HANGING AND SUPPORT COMPONENTS AND METHODS TO BE IN ACCORDANCE WITH ANSI/MSS SP-58-2009.

F. FOR INSULATED PIPING PROVIDE PIPE HANGER INSULATION SHIELD ACCORDING TO HANGER MANUFACTURER.

G. HANGER SPACING DOES NOT APPLY TO UNDERGROUND AND BELOW SLAB PIPING.

H. DUCTILE IRON AND CAST IRON PIPING MUST HAVE POLYETHYLENE ENCASEMENT WHEN INSTALLED IN CORROSIVE SOILS, VERIFY IN FIELD.

I. HUBLESS CAST IRON JOINTS MUST BE MADE WITH ASTM C1540 HEAVY DUTY SHIELDED COUPLINGS WITH CENTER STOPS.

J. PVC PIPE MUST BE SOLID WALL, FOAM CORE IS NOT PERMITTED.

K. DO NOT USE PVC PIPING IN RETURN AIR PLENUM.

**NOTES:**

1. AT LEAST 1 ADDITIONAL HANGER IS REQUIRED AT EACH JOINT.
2. 10' HANGER SPACING IS PERMITTED ONLY IF 10' LENGTHS OF PIPE ARE USED.
3. PIPE MATERIAL MAY BE USED WHERE PERMITTED BY AUTHORITY HAVING JURISDICTION.
4. FOR SIZES 2 INCHES AND SMALLER, A GUIDE SHALL BE INSTALLED MIDWAY BETWEEN REQUIRED VERTICAL SUPPORTS. SUCH GUIDES SHALL PREVENT PIPE MOVEMENT IN A DIRECTION PERPENDICULAR TO THE AXIS OF THE PIPE.

# MINIMUM PIPE INSULATION THICKNESS

| FLUID OPERATING TEMPERATURE RANGE (°F) | INSULATION CONDUCTIVITY | | NOMINAL PIPE OR TUBE SIZE (IN) | | | | |
|---|---|---|---|---|---|---|---|
| | CONDUCTIVITY BTU*IN./(H*FT2**F) | MEAN RATING TEMPERATURE, °F | < 1 | 1 TO < 1-1/2 | 1-1/2 TO < 4 | 4 TO < 8 | ≥ 8 |
| > 350 | 0.32 - 0.34 | 250 | 4.5 | 5.0 | 5.0 | 5.0 | 5.0 |
| 251-350 | 0.29 - 0.32 | 200 | 3.0 | 4.0 | 4.5 | 4.5 | 4.5 |
| 201-250 | 0.27 - 0.30 | 150 | 2.5 | 2.5 | 2.5 | 3.0 | 3.0 |
| 141-200 | 0.25 - 0.29 | 125 | 1.5 | 1.5 | 2.0 | 2.0 | 2.0 |
| 105-140 | 0.21 - 0.28 | 100 | 1.0 | 1.0 | 1.5 | 1.5 | 1.5 |
| 61-104 | - | 82 | 0.5 | 0.5 | 1.0 | 1.0 | 1.0 |
| 40-60 | 0.21 - 0.27 | 75 | 0.5 | 0.5 | 1.0 | 1.0 | 1.0 |
| < 40 | 0.20 - 0.26 | 50 | 0.5 | 1.0 | 1.0 | 1.0 | 1.0 |

**NOTES:**

1. ALL PIPING TO THE INLET OF A WATER HEATER AND PIPING CONVEYING WATER HEATED BY A WATER HEATER SHALL BE INSULATED. DOMESTIC COLD WATER SHALL BE INSULATED IN ACCORDANCE WITH THE TABLE ABOVE BASED ON FLUID TEMPERATURE DEPENDING ON GROUNDWATER AT PROJECT LOCATION.
2. FOR PIPING SMALLER THAN 1-1/2 INCHES AND LOCATED IN PARTITIONS WITHIN CONDITIONED SPACES, REDUCTION OF THESE THICKNESSES BY 1 INCH SHALL BE PERMITTED (BEFORE THICKNESS ADJUSTMENT REQUIRED IN FOOTNOTE B) BUT NOT TO A THICKNESS LESS THAN 1 INCH.
3. MISCELLANEOUS FLUIDS SUCH AS WASTE PIPING FROM ELECTRIC WATER COOLERS TO DRAINAGE, REFRIGERANT SUCTION LINES OR HOT GAS BYPASS LINES, AND COOLING COIL CONDENSATE PIPING ARE TO BE INSULATED IN ACCORDANCE WITH THE TABLE ABOVE BASED ON FLUID TEMPERATURE.

# WATER HEATER SCHEDULE

| MARK | PRODUCT | | TANK CAPACITY | ELECTRICAL | | | WATER | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | MANUFACTURER | MODEL | | KW | FLA | VOLT / PH / HZ | IN/OUT SIZE | DRAIN SIZE | DELTA T | TEMP. |
| EWH | LOCHINVAR | LDJ-10-JE | 10 GAL | 2.5 KW | - | 208/1/60 | 3/4" | 3/4" | 78°F | 140°F |

NOTES: 62°F COLD WATER TEMPERATURE USED FOR CALCULATIONS

1. PROVIDE A 2 GALLON BLADDER EXPANSION TANK (WATTS PLT-5-M1 OR EQUAL) ON COLD WATER INLET, INSTALL DOWNSTREAM OF CHECK VALVE.
2. PROVIDE AUTOMATIC LEAK DETECTOR SHUTOFF SYSTEM (FLOODSTOP FS3/4NPT V4 OR EQUAL) ON DOMESTIC COLD WATER SUPPLY TO WATER HEATER. PROVIDE SENSOR LOCATED IN WATER HEATER DRAIN PAN.
   A. PROVIDE LEAK DETECTION CONTROL PANEL WITH PLUG-IN POWER CONNECTION. PROVIDE 120V RECEPTACLE AT WATER HEATER LOCATION FOR CONNECTION. REFER TO ELECTRICAL PLANS FOR RECEPTACLE AND CIRCUIT REQUIREMENTS.
   B. SYSTEM SHALL AUTOMATICALLY SHUT OFF WATER UPON LEAK DETECTION AND REQUIRE MANUAL RESET.
3. PROVIDE HEAT TRAPS ON HOT AND COLD WATER CONNECTIONS.
4. ROUTE WATER HEATER DRAIN PAN TO NEAREST FLOOR DRAIN OR MOP SINK WITH AIR GAP.

# CIRCULATION PUMP SCHEDULE

| MARK | SERVICE | GPM | HEAD (FT.) | MOTOR | V/PH/HZ | AMPS | RPM | BASIS OF DESIGN (OR EQUAL) | NOTES |
|---|---|---|---|---|---|---|---|---|---|
| CP | DOMESTIC HOT WATER | - | - | 1 / 40 | 115/1/60 | 0.52 | 3250 | TACO 006-ST4 | ALL |

**PUMP OPERATION NOTES:**

1. PROVIDE RECIRCULATION PUMP WITH INTEGRAL TIMER; PROVIDE SEPERATE TIMER IF NOT INCLUDED WITH PUMP. TIMER SHALL PREVENT PUMP OPERATION FOR A MINIMUM OF 20 MINUTES (ADJ.) AFTER SHUTDOWN.
2. INSTALL AQUASTAT ON HOT WATER RETURN LINE AND CONTROL PUMP OPERATION AS FOLLOWS:
   A. ACTIVATE PUMP WHEN RETURN WATER TEMPERATURE DROPS BELOW 100°F (ADJ.)
   B. DEACTIVATE PUMP WHEN RETURN WATER TEMPERATURE EXCEEDS 120°F (ADJ.)
3. PROVIDE BALANCING VALVE IN RECIRCULATION LINE AND SET FLOW TO 0.5 GPM.
4. PROVIDE CHECK VALVE IN RECIRCULATION LINE DOWNSTREAM OF PUMP OR UTILIZE PUMP WITH INTEGRAL CHECK VALVE.
5. PROVIDE 115V, SINGLE-PHASE POWER CONNECTION AT PUMP LOCATION. REFER TO ELECTRICAL PLANS FOR CIRCUITING AND CONNECTION REQUIREMENTS.

# PIPE MATERIALS

| CONDITION/LOCATION | MATERIAL TYPE |
|---|---|
| ABOVE GROUND WATER | PEX-A[1] |
| BELOW GROUND WATER | PEX-A[2] |
| WASTE (UNDER 2") | SCHEDULE 40 PVC PLASTIC |
| WASTE (ABOVE 2") | SCHEDULE 40 PVC PLASTIC |
| WASTE | SCHEDULE 40 PCV PLASTIC |
| VENT | SCHEDULE 40 PCV PLASTIC |
| VENT (UNDER 2") | SCHEDULE 40 PVC PLASTIC |
| VENT (ABOVE 2") | CAST IRON |
| CONDENSATE PIPING | SCHEDULE 40 PVC PLASTIC |

**NOTE:** IF THE ABOVE MATERIALS ARE NOT PERMITTED BY LOCAL CODES, USE THE FOLLOWING PIPING MATERIALS:
- TYPE "L" COPPER FOR ABOVE GROUND WATER LINES
- TYPE "K" COPPER FOR BELOW GROUND WATER LINES
- TYPE "M" COPPER FOR WASTE AND VENT LINES UNDER 2"
- CAST IRON FOR WASTE LINES 2" AND LARGER
- SCHEDULE 40 WROUGHT IRON FOR CONDENSATE PIPING

1. USE TYPE "L" COPPER WHERE PIPING IS EXPOSED TO UV LIGHT.

2. USE TYPE "K" COPPER WHERE PIPING IS EXPOSED TO UV LIGHT.

# SLOPE OF HORIZONTAL DRAINAGE PIPE

| SIZE(INCHES) | MINIMUM SLOPE (INCHES PER FOOT) |
|---|---|
| 2-1/2 OR LESS | 1/4 |
| 3 TO 6 | 1/8 |
| 8 OR LARGER | 1/16 |

**NOTE: TABLE INFORMATION FROM IPC TABLE 704.1**

# MAXIMUM DISTANCE OF FIXTURE TRAP FROM VENT

| SIZE PR TRAP (INCHES) | SLOPE (INCHES PER FOOT) | DISTANCE FROM TRAP (FEET) |
|---|---|---|
| 1-1/4 | 1/4 | 5 |
| 1-1/2 | 1/4 | 6 |
| 2 | 1/4 | 8 |
| 3 | 1/8 | 12 |
| 4 | 1/8 | 16 |

**NOTE: TABLE INFORMATION FROM IPC TABLE 909.1**

# MINIMUM SIZE OF FIXURE WATER SUPPLY PIPES

| FIXTURE (1) | MIN. PIPE SIZE (INCH) |
|---|---|
| DISHWASHER, DOMESTIC (2) | 1/2 |
| DRINKING FOUNTAIN | 3/8 |
| HOSE BIBB | 1/2 |
| KITCHEN SINK (2) | 1/2 |
| LAVATORY | 3/8 |
| SINKS, SERVICE | 1/2 |
| URINAL, FLUSH TANK | 1/2 |
| URINAL, FLUSHOMETER VALVE | 3/4 |
| WALL HYDRANT | 1/2 |
| WATER CLOSET, FLUSH TANK | 3/8 |
| WATER CLOSET. FLUSHOMETER TANK | 3/8 |
| WATER CLOSET. FLUSHOMETER VALVE | 1 |
| WATER CLOSET. FONE PIECE (2) | 1/2 |

**NOTES:**

1. TABLE INFORMATION GATHERED FROM IPC TABLE 604.5.

2. WHERE THE DEVELOPED LENGTH OF THE DISTRIBUTION LINE IS 50 FEET OR LESS, AND THE AVAILABLE PRESSURE AT THE METER IS 35 PSI OR GREATER, THE MINIMUM SIZE OF AN INDIVIDUAL DISTRIBUTION LINE SUPPLIED FROM A MANIFOLD AND INSTALLED AS PART OF A PARALLEL WATER DISTRIBUTION SYSTEM SHALL BE ONE NOMINAL TUBE SIZE SMALLER THAN THE SIZES INDICATED.

# PIPING VOLUME AND MAXIMUM PIPING LENGTHS

| PIPING SIZE (INCHES) | VOLUME (OZ/FT) | MAX. PIPING LENGTH (FEET) | |
|---|---|---|---|
| | | PUBLIC LAVATORY | OTHER FIXTURES |
| 1/4 | 0.33 | 6 | 50 |
| 5/16 | 0.5 | 4 | 50 |
| 3/8 | 0.75 | 3 | 50 |
| 1/2 | 1.5 | 2 | 43 |
| 5/8 | 2 | 1 | 32 |
| 3/4 | 3 | 0.5 | 21 |

**NOTE: TABLE INFORMATION FROM IECC TABLE C404.5.1**

# PLUMBING FIXTURE SCHEDULE

| MARK | DESCRIPTION | BASIS OF DESIGN | TRIM/ACCESSORIES | CONNECTION SIZES | | | | NOTES |
|---|---|---|---|---|---|---|---|---|
| | | | | CW | HW | WASTE | VENT | |
| SK-T | TOP-MOUNTED LOUNGE SINK | BELAY LRAD-2219 | SINK DRAIN OFFSET TO BACK, PROVIDE WITH LK-35 STRAINER, MCGUIRE #1912 P-TRAP AND # 165 STOPS WITH SUPPLIES. | 1/2" | 1/2" | 2" | 1-1/2" | 1, 2, 3, 4 |
| | LOUNGE FAUCET | SYMMONS DIA S-3510-PD | 1.0 GPM FAUCET, PROVIDE WITH KOHLER K-18036 0.5GPM FEMALE AERATOR. | 3/8" | 3/8" | - | - | 3 |
| SK-U | UNDER-MOUNT LIBRARY SINK | ELKAY DCFU1618 | SINGLE BOWL CENTERED DRAIN OFFSET TO BACK, PROVIDE WITH LK-35 STRAINER, MCGUIRE #1912 P-TRAP AND #165 STOPS WITH SUPPLIES. | 1/2" | 1/2" | 2" | 1-1/2" | 1, 2, 3, 4 |
| | LIBRARY FAUCET | SYMMONS DIA S-3510-PD | 1.0 GPM FAUCET, PROVIDE WITH KOHLER K-18036 0.5GPM FEMALE AERATOR. | 3/8" | 3/8" | - | - | 3 |
| WC | WATER CLOSET - ADA COMPLIANT | KOHLER K-3519-T-0 | FLOOR MOUNTED (1.0 GPF) VITREOUS CHINA, ELONGATED BOWL, AND TANK COVER LOCKS. PROVIDE BEMIS 1655SSCT SEAT WITH SELF-SUSTAINING STAINLESS STEEL CHECK HINGE. PROVIDE KOHLER K-7637 3/8" ANGLE SUPPLY WITH STOP AND ESCUTCHEON. | 1" | - | 4" | 2" | 1, 3 |
| LAV | LAVATORY -ADA COMPLIANT | KOHLER K-2005 COMPLETE | WALL HUNG ADA COMPLIANT LAVATORY, PROVIDE WITH K7715 OUTLET WITH TAILPIECE, ZURN Z1231 FIXTURE SUPPORT, MCGUIRE ST02LK SUPPLIES WITH STOPS AND MCGUIRE #8902 P-TRAP. INSULATE P-TRAP, STOPS AND SUPPLIES WITH "PRO-WRAP" BY MCGUIRE. MOUNT WITH RIM MAXIMUM 34" AFF. | 1/2" | 1/2" | 2" | 1-1/2" | 1, 2, 3, 4 |
| | LAVATORY - (BATTERY) FAUCET | ZURN AQUASENSE Z6915-XL | BATTERY SENSOR ACTIVATED FAUCET WITH 0.5 GPM SPOUT AERATOR. PROVIDE WITH KOHLER K-7129-A DRAIN AND SYMMONS 2-225-CK-MS MIXING BELOW SINK, ADJUSTED TO PROVIDE 105F TO HOT WATER SIDE OF FAUCET. | 3/8" | 3/8" | - | - | 2, 3 |
| EDF | WATER COOLER - HILOW ADA COMPLIANT | ELKAY EZSTL8C | BI-LEVEL, STAINLESS STEEL CABINET, WITH WATERWAYS MANUFACTURED OF 100% LEAD FREE MATERIAL, ZURN Z1225 FIXTURE SUPPORT, BALL VALVE STOP WITH SUPPLY, SAFETY-GUARD BUBBLER. MCGUIRE #8872 P-TRAP. FULLY INSULATE P-TRAP. MOUNT WITH LOWER SPOUT OUTLET 36" ABOVE FINISH FLOOR. PROVIDE COLOR CHART FOR ARCHITECT COLOR SELECTION. PROVIDE WITH ELKAY MODEL #LKAPREZL CANE APRON AS REQUIRED. | 1/2" | - | 2" | 1-1/2" | 1, 2, 3 |
| MS | MOP SINK BASIN | FIAT MSK-2424 | 24"X24"X10" FIBERGLASS OR SIMILAR AND E77-AA BUMPER GUARD. T-35 HOSE WITH WALL HOOK, STAINLESS STEEL BACKSPLASH AND KOHLER K-8907 WITH VACUUM BREAKER. REFER TO ARCHITECTURAL ELEVAITONS FOR FRP HEIGHTS. | - | - | 3" | 1-1/2" | 3 |
| | WALL FAUCET | KOHLER K-837T60-4-CP | TRITON BOWE SERVICE SINK FAUCET, WALL-TO-SPOUT OUTLET WITH WALL BRACE AND PAIL HOOK. | 1/2" | 1/2" | - | - | 3 |
| FL | FLOOR DRAIN | J.R. SMITH #2010 | 6" ROUND NICKEL BRONZE GRATE. PROVIDE WITH TRAP PRIMER CONNECTION. PROVIDE WITH J.R. SMITH TRAP INSERT. | - | - | 3" | - | N/A |
| DB | DRAIN BOX | SIOUX CHIEF MODEL #696-3F | PROVIDE DRAIN BOX WITH #696-LC LOUVERED COVER, #696-CF SECONDARY DRAINAGE FUNNEL, AND J.R. SMITH TRAP SEAL INSERT. BOX TO COME COMPLETE WITH WALL FLANGE AND LOUVER. PIPE INDIRECT IN WALL TO DRAIN BOX. COORDINATE EXACT LOCATION WITH ARCHITECT AND GC. | - | - | 2" | - | N/A |
| WH | WALL HYDRANT | J.R. SMITH #5509-QT | PROVIDE WITH INTEGRAL BACKFLOW PREVENTER, LATCHING COVER, FREEZE-PROOF AND OF PROPER LENGTH FOR WALL IN WHICH INSTALLED, ALL BRONZE BOX. VALVE SEAT MUST BE ON BUILDING SIDE OF EXTERIOR WALL INSULATION. INSTALL WITH CENTER LINE 24" ABOVE FINISH GRADE. PROVIDE OWNER WITH ONE (1) LOOSE KEY FOR EACH WALL HYDRANT. | 3/4" | - | - | - | N/A |
| RH | ROOF HYDRANT | JR SMITH #5906-H | PROVIDE WITH DRAIN TUBE PIPED TO NEAREST DRAIN. DRAIN PIPE TO BE COPPER TO THE NEAREST DRAIN OR JANITORS RECEPTOR. | 3/4" | - | 3/8" | - | N/A |
| RD | ROOF DRAIN | J.R. SMITH #1011 | COMPLETE WITH SUMP RECEIVER AND UNDER DECK CLAMP, AND CAST IRON OR ALUMINUM DOME. PROVIDE 4" HIGH, 1/16: THICK PERFORATED STAINLESS STEEL GRAVEL TRAP AROUND DOME. | - | - | SEE PLAN | - | N/A |
| OD | OVERFLOW DRAIN | J.R. SMITH #1080 | COMPLETE WITH SUMP RECEIVER, UNDER DECK CLAMP, 2" EXTERNAL DAM, AND CAST IRON OR ALUMINUM DOME. | - | - | SEE PLAN | - | N/A |
| CD | CANOPY DRAIN | ZURN ZRB185 | 6" DIAMETER CORNICE DRAIN DURA-COATED CAST IRON, SIDE OUTLET BODY, PLAIN BRONZE DOME AND MEMBRANE FLASHING CLAMP. | - | - | SEE PLAN | - | N/A |
| DSN | DOWNSPOUT NOZZLE | J.R. SMITH #1770 | DOWNSPOUT NOZZLE. MOUNT 1'-0" ABV FINISH GRADE. COORDINATE EXACT LOCATION WITH ARCHITECT. ENSURE PROTRUSION IS UNDER 4" FOR ADA CLEARANCE. | - | - | SEE PLAN | - | N/A |
| VB | WALL VALVE BOX | OATEY 38689 | STEEL WALL MOUNTED BOX WITH 1/4 TURN BRASS BALL VALVE FOR CONNECTION TO COUTERTOP DRINKING EQUIPMENT. | 1/2" | - | - | - | N/A |

**NOTES:**

1. REFER TO ARCHITECTURAL ELEVATIONS FOR MOUNTING HEIGHT AND ARCHITECTURAL DRAWINGS FOR EXACT MOUNTING POSITION OF FIXTURES.
2. FAUCET HOLES TO MATCH FAUCET SPECIFIED.
3. FIXTURE ASSEMBLY MUST BE APPROVED BY AND INSTALLED PER ADA REQUIREMENTS.
4. PROVIDE INSULATION KIT ON ALL ADA FIXTURES EXPOSED SUPPLIES AND DRAIN TRAP WITH TRUBRO LAV-GUARD IN WHITE.

* BASIS OF DESIGN PRODUCTS ARE INDICATED FOR REFERENCE ONLY. COMPARABLE PRODUCTS WILL BE ACCEPTED PROVIDED COMPLIANCE WITH TECHNICAL SPECIFICATION REQUIREMENTS.

# CLEANOUT SCHEDULE

| MARK | DESCRIPTION | BASIS OF DESIGN | TRIM/ACCESSORIES | DRAIN |
|---|---|---|---|---|
| FCO | CLEAN-OUT (FLOOR) | J.R. SMITH #4023S(INSIDE) #4240 (OUTSIDE) | INTERIOR 3/4" DIA, MEDIUM-DUTY CAST IRON FLOOR MOUNTED CLEANOUT WITH ROUND ADJUSTABLE SCORIATED SECURED NICKEL BRONZE TOP. TAPER THREAD. PROVIDE PVC PLUG WHERE PERMITTED BY LOCAL CODES; OTHERWISE, PROVIDE BRONZE PLUG. | PER PLAN |
| WCO | CLEAN-OUT (WALL) | J.R. SMITH #4422C | WALL MOUNTED CLEANOUT WITH ACCESS COVER, TAPER THREAD. PROVIDE PVC PLUG WHERE PERMITTED BY LOCAL CODES; OTHERWISE, PROVIDE BRONZE PLUG. | PER PLAN |

* BASIS OF DESIGN PRODUCTS ARE INDICATED FOR REFERENCE ONLY. COMPARABLE PRODUCTS WILL BE ACCEPTED PROVIDED COMPLIANCE WITH TECHNICAL SPECIFICATION REQUIREMENTS.


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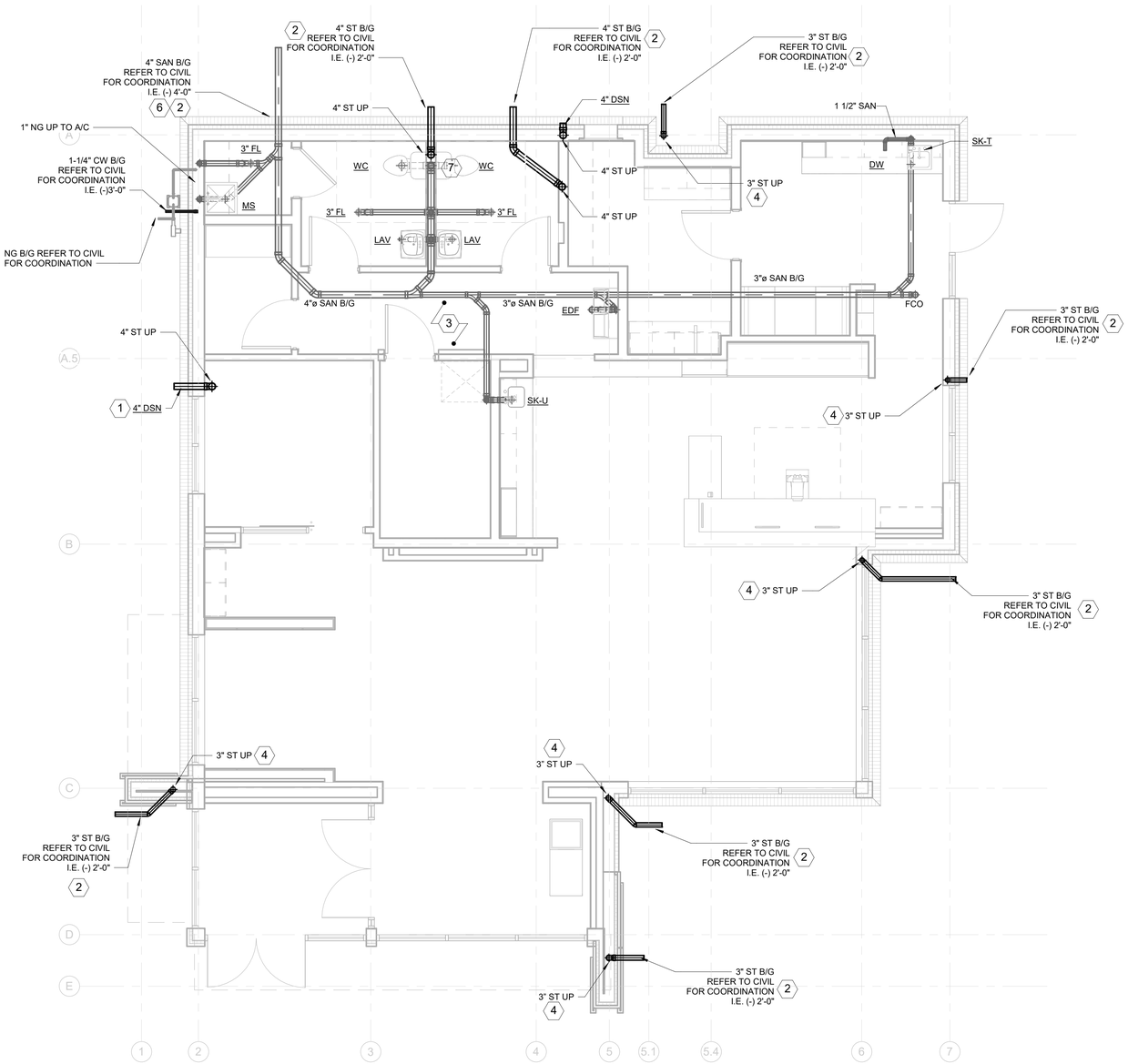
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Sheet Title:
**PLUMBING TABLES AND SCHEDULES**

Sheet Number:
**P-101**

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1 PLUMBING PROPOSED FOUNDATION PLAN
P-301 1/4" = 1'-0"

MECHANICAL GENERAL NOTES
(APPLIES TO THIS SHEET ONLY)

1. LOCATIONS OF UTILITIES SHOWN ON PLANS ARE APPROXIMATE, VERIFY WITH LOCAL UTILITY PRIOR TO BIDDING.
2. CONTRACTOR SHALL VERIFY EXACT LOCATION, SIZE AND ELEVATION OF ALL EXISTING SERVICES PRIOR TO INSTALLING ANY NEW PIPE.
3. ALL OUTSIDE CLEANOUTS SHALL BE BROUGHT TO GRADE AND EMBEDDED\ IN 18"X18"X6" THICK CONCRETE PAD. (J.R. SMITH 4258 OR EQUAL.)
4. WHEREVER DISSIMILAR METALS ARE CONNECTED ON WATER LINES, A DIELECTRIC UNION SHALL BE USED.
5. ALL HORIZONTAL WATER AND VENT PIPING SHALL BE RUN ABOVE CEILING ON PLAN WHICH SHOWN UNLESS OTHERWISE NOTED.
6. ALL HORIZONTAL SANITARY PIPING IS RUN BELOW FLOOR ON PLAN WHERE SHOWN UNLESS OTHERWISE NOTED.
7. ALL WATER PIPING BELOW SLAB ON GRADE SHALL BE BENT UP AT ENDS SO THAT NO JOINTS OCCUR BELOW FLOOR.
8. COORDINATE ALL PIPE ROUTING TO AVOID CONFLICTS WITH STRUCTURAL, MECHANICAL, AND ELECTRICAL FEATURES OF BUILDING.
9. ALL WALL HYDRANTS AND HOSE BIBBS SHALL BE MOUNTED 24" ABOVE FINISH GRADE OR FINISH FLOOR UNLESS OTHERWISE NOTED.
10. ALL WATER PIPING INSTALLED IN EXTERIOR WALLS SHALL BE LOCATED ON THE INTERIOR SIDE OF THE EXTERIOR WALL INSULATION.
11. NO VENT THRU ROOF IS TO BE LOCATED WITHIN 10 FEET OF ANY BUILDING AIR INTAKES, PER CODE. COORDINATE WITH MECHANICAL AND GENERAL CONTRACTORS.
12. DOMESTIC WATER PIPING AND FIRE PROTECTION PIPING LOCATED ABOVE THE CEILING, SHALL BE INSTALLED BELOW CEILING INSULATION.
13. CONTRACTOR SHALL COORDINATE MECHANICAL FLOOR DRAIN LOCATIONS WITH MECHANICAL EQUIPMENT PRIOR TO INSTALLATION.
14. CONTRACTOR SHALL PROVIDE SHOCK ARRESTORS ON ALL BRANCH LINES.
15. CONTRACTOR SHALL COORDINATE ALL SINKS WITH CASEWORK PRIOR TO ORDERING SINKS.
16. ALL HOT WATER LINES TO BE INSULATED.

MECHANICAL KEY NOTES
(APPLIES TO THIS SHEET ONLY)

1. TERMINATE EMERGENCY DOWNSPOUT 12" ABOVE FINISH CURB. COORDINATE WITH ARCHITECT.
2. FOR CONTINUATION SEE CIVIL SITE UTILITY PLAN.
3. ALL UNDERGROUND SANITARY LINES ARE TO BE NO LESS THAN 3" IN SIZE.
4. VERTICAL STORM RISER TO DROP UNDERGROUND AND ROUTE OUT BUILDING AS INDICATED.
5. PIPE SHALL BE INSTALLED WITH A HORIZONTAL INTERCEPTING TRAP UNDERNEATH EXTERNAL WALL. CLEANOUT TO BE LEFT AS AN AIR INLET.
6. REFER TO CIVIL DRAWINGS FOR PROPOSED EXTERIOR CLEANOUT LOCATION, COORDINATE TO AVOID SIDEWALK PATH OR TRANSFORMER PAD.
7. PLUMBING CONTRACTOR TO ENSURE THAT A "DOUBLE COMBO WYE" FITTING IS USED AT THE LOW FLOW WATER CLOSET CONNECTION TO SANITARY MAIN IN AN EFFORT TO IMPROVE WASTE FLOW.

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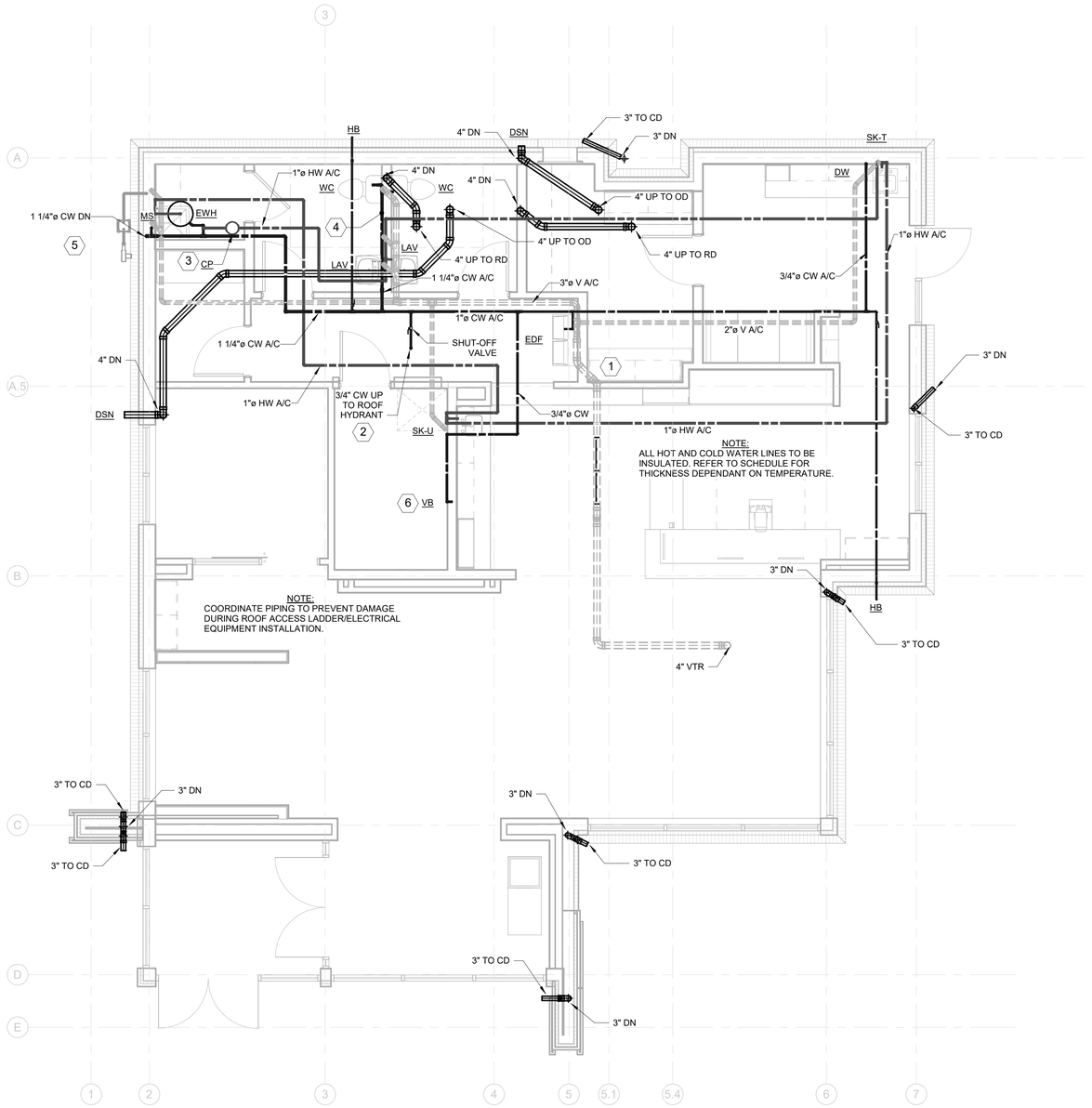
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Key Plan:

Sheet Title:
**PLUMBING SANITARY FOUNDATION PLAN**

Sheet Number:
**P-301**

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1 P-302 PLUMBING PROPOSED FLOOR PLAN
1/4" = 1'-0"

MECHANICAL GENERAL NOTES
(APPLIES TO THIS SHEET ONLY)

1. LOCATIONS OF UTILITIES SHOWN ON PLANS ARE APPROXIMATE, VERIFY WITH LOCAL UTILITY PRIOR TO BIDDING.
2. CONTRACTOR SHALL VERIFY EXACT LOCATION, SIZE AND ELEVATION OF ALL EXISTING SERVICES PRIOR TO INSTALLING ANY NEW PIPE.
3. ALL OUTSIDE CLEANOUTS SHALL BE BROUGHT TO GRADE AND EMBEDDED\ IN 18"X18"X6" THICK CONCRETE PAD. (J.R. SMITH 4258 OR EQUAL.)
4. WHEREVER DISSIMILAR METALS ARE CONNECTED ON WATER LINES, A DIELECTRIC UNION SHALL BE USED.
5. ALL HORIZONTAL WATER AND VENT PIPING SHALL BE RUN ABOVE CEILING ON PLAN WHICH SHOWN UNLESS OTHERWISE NOTED.
6. ALL HORIZONTAL SANITARY PIPING IS RUN BELOW FLOOR ON PLAN WHERE SHOWN UNLESS OTHERWISE NOTED.
7. ALL WATER PIPING BELOW SLAB ON GRADE SHALL BE BENT UP AT ENDS SO THAT NO JOINTS OCCUR BELOW FLOOR.
8. COORDINATE ALL PIPE ROUTING TO AVOID CONFLICTS WITH STRUCTURAL, MECHANICAL, AND ELECTRICAL FEATURES OF BUILDING.
9. ALL WALL HYDRANTS AND HOSE BIBBS SHALL BE MOUNTED 24" ABOVE FINISH GRADE OR FINISH FLOOR UNLESS OTHERWISE NOTED.
10. ALL WATER PIPING INSTALLED IN EXTERIOR WALLS SHALL BE LOCATED ON THE INTERIOR SIDE OF THE EXTERIOR WALL INSULATION.
11. NO VENT THRU ROOF IS TO BE LOCATED WITHIN 10 FEET OF ANY BUILDING AIR INTAKES, PER CODE. COORDINATE WITH MECHANICAL AND GENERAL CONTRACTORS.
12. DOMESTIC WATER PIPING AND FIRE PROTECTION PIPING LOCATED ABOVE THE CEILING, SHALL BE INSTALLED BELOW CEILING INSULATION.
13. CONTRACTOR SHALL COORDINATE MECHANICAL FLOOR DRAIN LOCATIONS WITH MECHANICAL EQUIPMENT PRIOR TO INSTALLATION.
14. CONTRACTOR SHALL PROVIDE SHOCK ARRESTORS ON ALL BRANCH LINES.
15. CONTRACTOR SHALL COORDINATE ALL SINKS WITH CASEWORK PRIOR TO ORDERING SINKS.

MECHANICAL KEY NOTES
(APPLIES TO THIS SHEET ONLY)

1. VENT PIPING IS TO RUN PARALLEL OVER SANITARY PIPING AND TIE INTO 3" VENT THROUGH ROOF.
2. DOMESTIC CW PIPING TO ROOF HYDRANT. FIXTURE REQUIRES 1/8" DRAIN LINE TO BE TIED INTO SANITARY.
3. SHUTOFF VALVES TO BE INSTALLED ON INPUT AND OUTPUT OF WATER HEATER IN A READILY ACCESSIBLE AND VISIBLE LOCATION, WITHIN 4'-0" TO 5'-0" AFF, UNOBSTRUCTED BY EQUIPMENT OR FINISH MATERIALS. A CIRCULATION PUMP AND EXPANSION TANK ARE TO BE INCLUDED WITH WATER HEATER INSTALLATION.
4. 1/2" SOFT COPPER BELOW SLAB TO PRIME FLOOR DRAIN. (TYPICAL)
5. INSTALL AT VERTICAL RISER HORIZONTAL PIPE SECTION FOR INSTALLATION OF A WHOLE BUILDING SEDIMENT FILTER ON THE INCOMING MAIN DOMESTIC WATER SUPPLY LINE. INSTALL A STANDARD FILTER SIMILAR TO "WHIRLPOOL LARGE CAPACITY 10" X 4.5" WHOLE HOUSE WATER FILTER SYSTEM WHKF-DWHBB" OR EQUIVALENT PRODUCT. PLUMBING CONTRACTOR TO INSTALL 10-MICROM FILTER FOR SYSTEM AND PROVIDE CLIENT WITH REPLACEMENT CARTIRIAGE.
6. TAP OFF THE INCOMING WATER SERVICE FROM THE SINK TO A WALL VALVE BOX, RUN A 1/4" FLEXIBLE COPPER OR PCV LINE TO THE FILTERS LOCATED IN THE QUENCH MACHEINE, FROM THERE RUN ANOTHER LINE TO THE KEURIG.

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Project Name:
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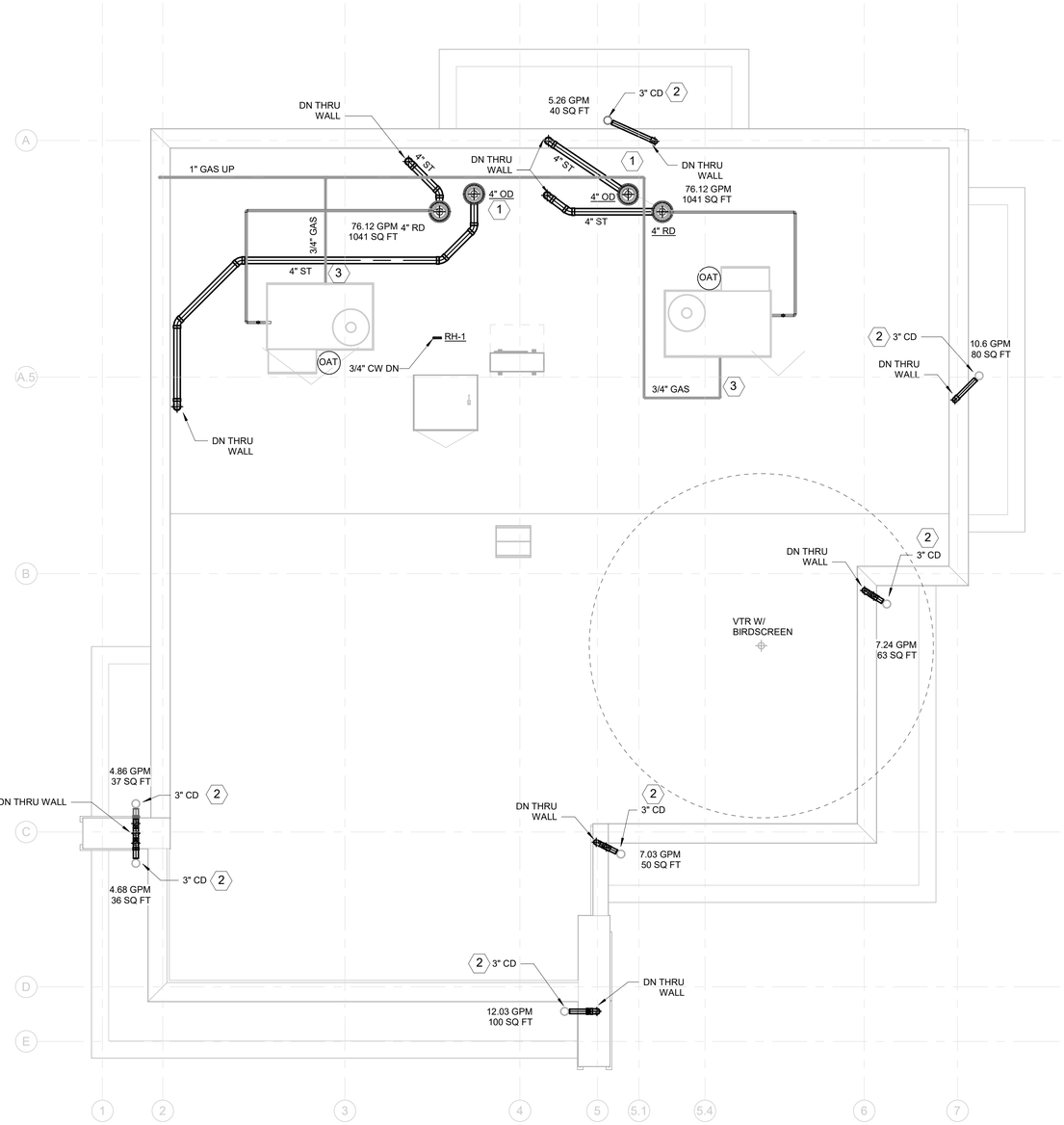
2711 WATSON BOULEVARD
WARNER ROBBINS, GA 31093

Key Plan:

Sheet Title:
**PLUMBING DOMESTIC FLOOR PLAN**

Sheet Number:
**P-302**

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1 P-303 PLUMBING STORM ROOF PLAN
1/4" = 1'-0"

MECHANICAL GENERAL NOTES
(APPLIES TO THIS SHEET ONLY)

1. LOCATIONS OF UTILITIES SHOWN ON PLANS ARE APPROXIMATE, VERIFY WITH LOCAL UTILITY PRIOR TO BIDDING.
2. CONTRACTOR SHALL VERIFY EXACT LOCATION, SIZE AND ELEVATION OF ALL EXISTING SERVICES PRIOR TO INSTALLING ANY NEW PIPE.
3. ALL OUTSIDE CLEANOUTS SHALL BE BROUGHT TO GRADE AND EMBEDDED\ IN 18"X18"X6" THICK CONCRETE PAD. (J.R. SMITH 4258 OR EQUAL.)
4. WHEREVER DISSIMILAR METALS ARE CONNECTED ON WATER LINES, A DIELECTRIC UNION SHALL BE USED.
5. ALL HORIZONTAL WATER AND VENT PIPING SHALL BE RUN ABOVE CEILING ON PLAN WHICH SHOWN UNLESS OTHERWISE NOTED.
6. ALL HORIZONTAL SANITARY PIPING IS RUN BELOW FLOOR ON PLAN WHERE SHOWN UNLESS OTHERWISE NOTED.
7. ALL WATER PIPING BELOW SLAB ON GRADE SHALL BE BENT UP AT ENDS SO THAT NO JOINTS OCCUR BELOW FLOOR.
8. COORDINATE ALL PIPE ROUTING TO AVOID CONFLICTS WITH STRUCTURAL, MECHANICAL, AND ELECTRICAL FEATURES OF BUILDING.
9. ALL WALL HYDRANTS AND HOSE BIBBS SHALL BE MOUNTED 24" ABOVE FINISH GRADE OR FINISH FLOOR UNLESS OTHERWISE NOTED.
10. ALL WATER PIPING INSTALLED IN EXTERIOR WALLS SHALL BE LOCATED ON THE INTERIOR SIDE OF THE EXTERIOR WALL INSULATION.
11. NO VENT THRU ROOF IS TO BE LOCATED WITHIN 10 FEET OF ANY BUILDING AIR INTAKES, PER CODE. COORDINATE WITH MECHANICAL AND GENERAL CONTRACTORS.
12. DOMESTIC WATER PIPING AND FIRE PROTECTION PIPING LOCATED ABOVE THE CEILING, SHALL BE INSTALLED BELOW CEILING INSULATION.
13. CONTRACTOR SHALL COORDINATE MECHANICAL FLOOR DRAIN LOCATIONS WITH MECHANICAL EQUIPMENT PRIOR TO INSTALLATION.
14. CONTRACTOR SHALL PROVIDE SHOCK ARRESTORS ON ALL BRANCH LINES.
15. CONTRACTOR SHALL COORDINATE ALL SINKS WITH CASEWORK PRIOR TO ORDERING SINKS.
16. ALL HOT WATER LINES TO BE INSULATED.

MECHANICAL KEY NOTES
(APPLIES TO THIS SHEET ONLY)

1. ENSURE OVERFLOW STORM DRAIN IS INSTALLED AT AN ELEVATION OF 2" HIGHER THAN THE PRIMARY STORM DRAIN.
2. 2 CANOPY STORM DRAINAGE CONNECTION WITH CIVIL. CANOPY STORM DRAINAGE SLOPE IS 1/8".
3. PENETRATE ROOF WITH NATURAL GAS PIPING AT EACH RTU. INSTALL A MANUAL SHUTOFF VALVE, DRIP LEG (SEDIMENT TRAP), TEST PORTS UPSTREAM AND DOWNSTREAM OF PRESSURE REGULATOR, LINE PRESSURE REGULATOR (SIZED AND ADJUSTED PER RTU REQUIREMENTS), UNION, AND FLEXIBLE GAS CONNECTOR REDUCED TO RTU INLET SIZE. SUPPORT ALL PIPING PER CODE. REFER TO GAS PIPING DETAIL.

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Client:
**FIFTH THIRD BANK**

Project Number: 26032
Project Name:
**FIFTH THIRD BANK WATSON 1,900 SF**

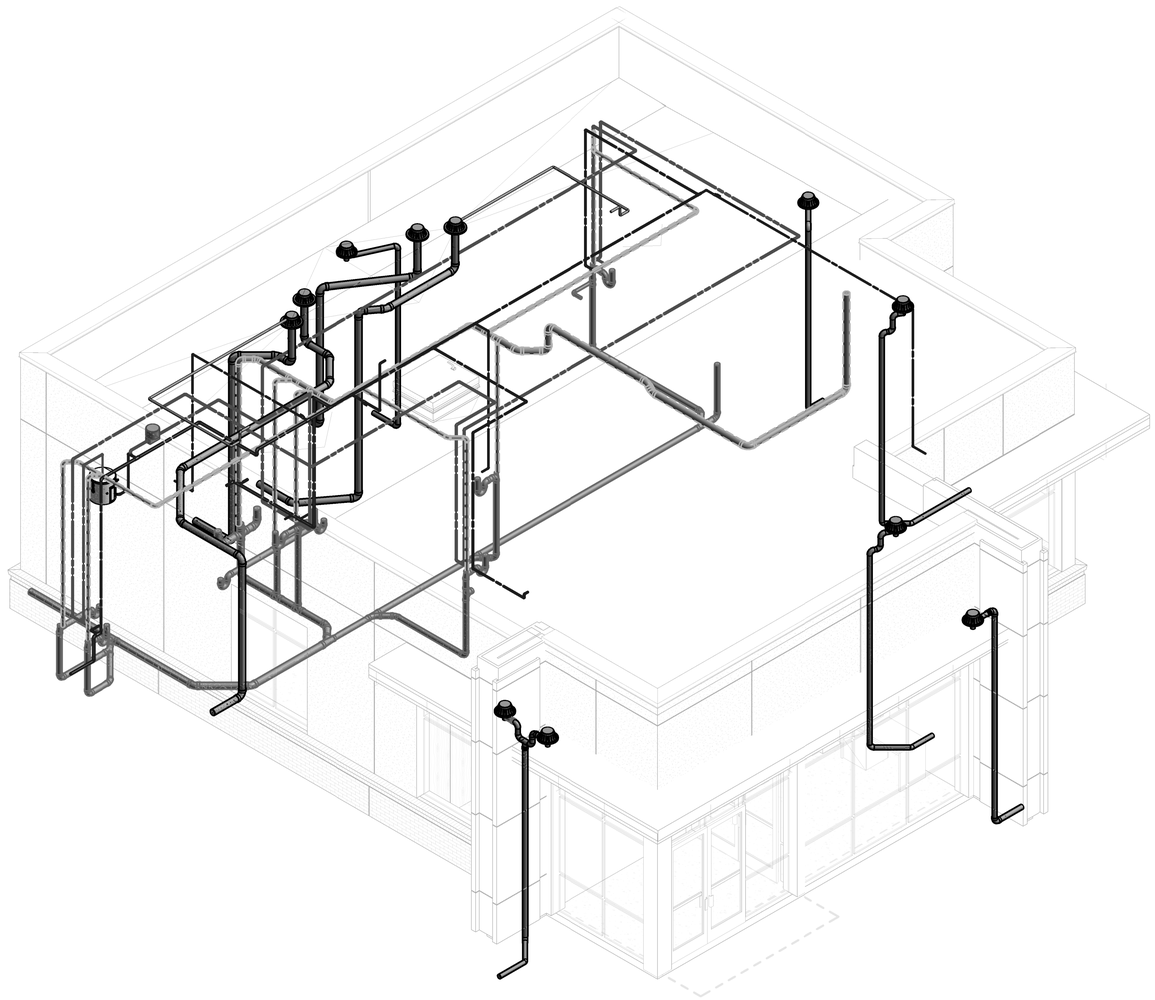
2711 WATSON BOULEVARD
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Key Plan:

Sheet Title:
**PLUMBING PROPOSED ROOF PLAN**

Sheet Number:
**P-303**

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1 P-401 PLUMBING ISO PLAN

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Client:

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Project Name:

**FIFTH THIRD BANK**
**WATSON**
**1,900 SF**

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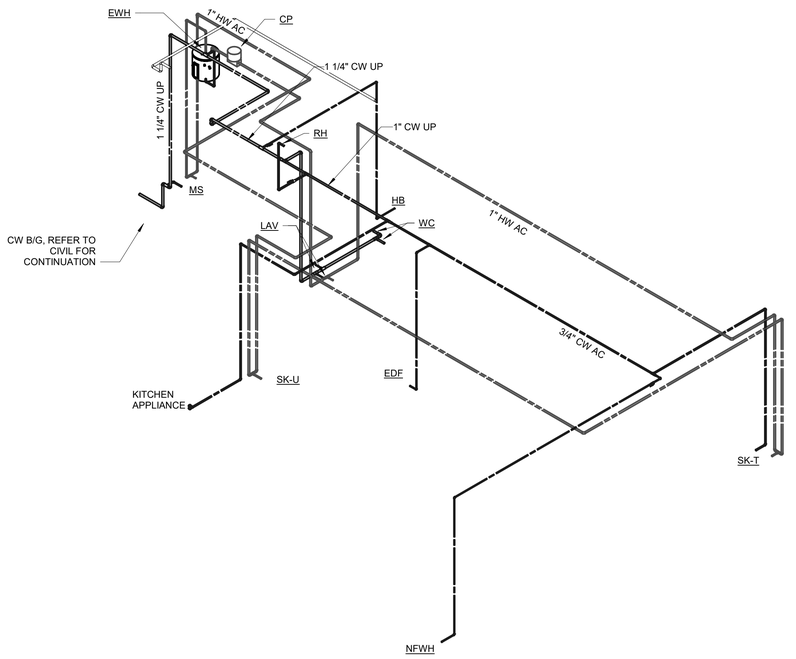
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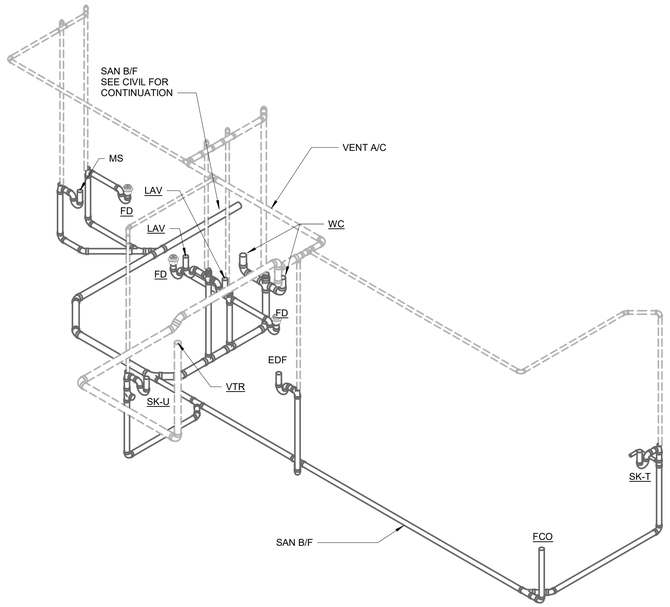
**PLUMBING BUILDING ISOMETRIC**

Sheet Number:

**P-401**




1 P-402 DOMESTIC WATER RISER




2 P-402 SAN/VENT RISER

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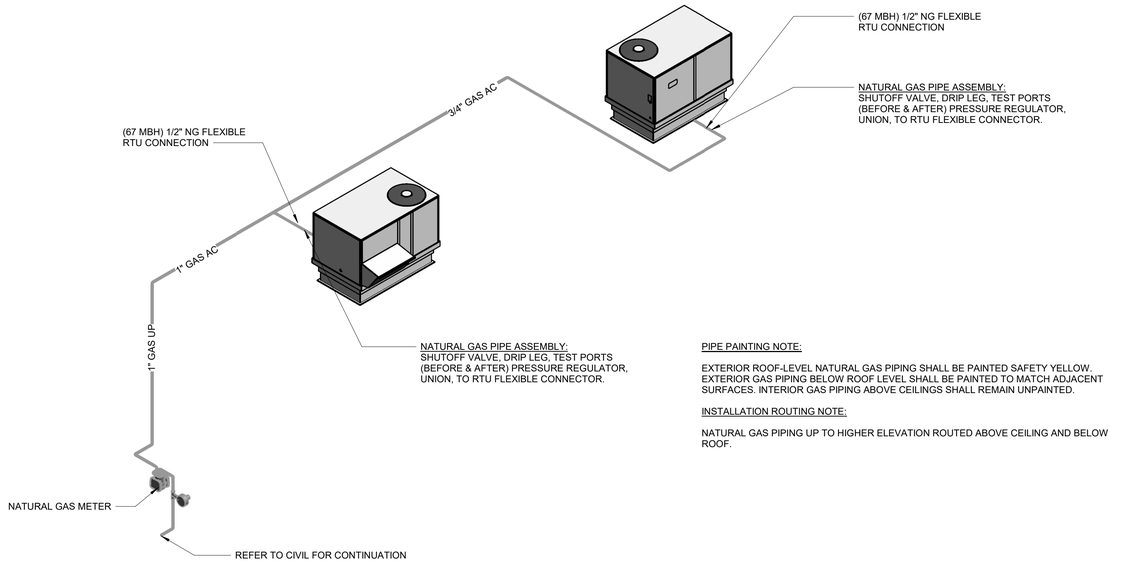
Project Number: 26032
Project Name:
**FIFTH THIRD BANK WATSON 1,900 SF**

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Key Plan:

Sheet Title:
**PLUMBING ISOMETRIC RISER DIAGRAM**

Sheet Number:
**P-402**



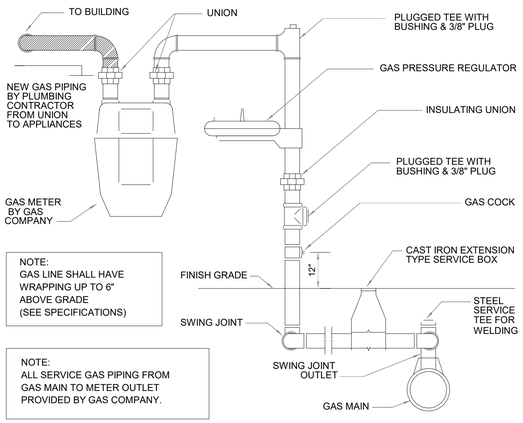

PIPE PAINTING NOTE:

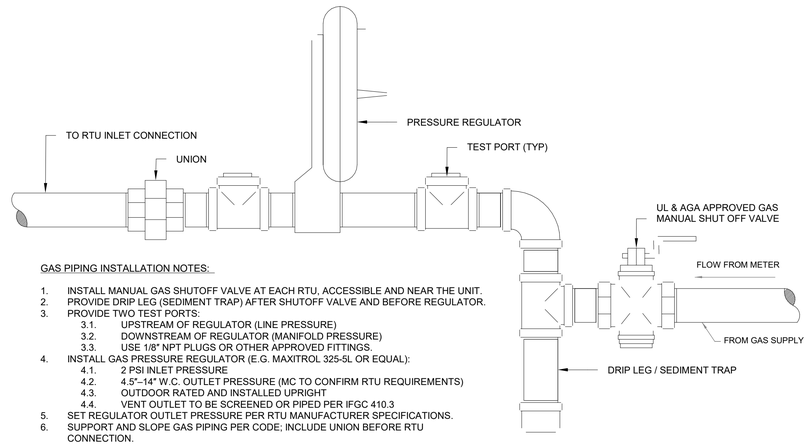
EXTERIOR ROOF-LEVEL NATURAL GAS PIPING SHALL BE PAINTED SAFETY YELLOW. EXTERIOR GAS PIPING BELOW ROOF LEVEL SHALL BE PAINTED TO MATCH ADJACENT SURFACES. INTERIOR GAS PIPING ABOVE CEILINGS SHALL REMAIN UNPAINTED.

INSTALLATION ROUTING NOTE:

NATURAL GAS PIPING UP TO HIGHER ELEVATION ROUTED ABOVE CEILING AND BELOW ROOF.

1 P-403 PLUMBING NATURAL GAS ISO PLAN




GAS PIPING INSTALLATION NOTES:

1. INSTALL MANUAL GAS SHUTOFF VALVE AT EACH RTU, ACCESSIBLE AND NEAR THE UNIT.
2. PROVIDE DRIP LEG (SEDIMENT TRAP) AFTER SHUTOFF VALVE AND BEFORE REGULATOR.
3. PROVIDE TWO TEST PORTS:
   - 3.1. UPSTREAM OF REGULATOR (LINE PRESSURE)
   - 3.2. DOWNSTREAM OF REGULATOR (MANIFOLD PRESSURE)
   - 3.3. USE 1/8" NPT PLUGS OR OTHER APPROVED FITTINGS.
4. INSTALL GAS PRESSURE REGULATOR (E.G. MAXITROL 325-5L OR EQUAL):
   - 4.1. 2 PSI INLET PRESSURE
   - 4.2. 4.5"–14" W.C. OUTLET PRESSURE (MC TO CONFIRM RTU REQUIREMENTS)
   - 4.3. OUTDOOR RATED AND INSTALLED UPRIGHT
   - 4.4. VENT OUTLET TO BE SCREENED OR PIPED PER IFGC 410.3
5. SET REGULATOR OUTLET PRESSURE PER RTU MANUFACTURER SPECIFICATIONS.
6. SUPPORT AND SLOPE GAS PIPING PER CODE; INCLUDE UNION BEFORE RTU CONNECTION.

**PIPE PAINTING NOTE:**

EXTERIOR ROOF-LEVEL NATURAL GAS PIPING SHALL BE PAINTED SAFETY YELLOW. EXTERIOR GAS PIPING BELOW ROOF LEVEL SHALL BE PAINTED TO MATCH ADJACENT SURFACES. INTERIOR GAS PIPING ABOVE CEILINGS SHALL REMAIN UNPAINTED.

**INSTALLATION ROUTING NOTE:**

NATURAL GAS PIPE UP TO HIGHER ELEVATION ROUTED ABOVE CEILING AND BELOW ROOF.

**NATURAL GAS PIPING CALCULATIONS AND NOTES:**

1. NATURAL GAS PIPING SHALL BE SCHEDULED 40 BLACK STEEL COMPLYING WITH ANSI B35.10. FITTINGS SHALL BE STEEL OR MALLEABLE IRON. JOINTS SHALL BE THREADED OR WELDED.
2. CONTRACTOR SHALL BE RESPONSIBLE FOR CONTACTING AND COORDINATING WITH THE LOCAL UTILITY COMPANIES (GAS, WATER, SEWER, ELECTRICAL, TELEPHONE) TO ENSURE TIMELY INSTALLATION OF UTILITY SERVICES TO THE POINTS SHOWN ON THE PLANS.
3. REFER TO THE ARCHITECTURAL PLANS FOR LOCATIONS OF ALL FIXTURES AND EQUIPMENT. PROVIDE ALL FINAL CONNECTIONS PER KITCHEN CONSULTANT SCHEDULES.
4. ALL GAS PIPING IN SCOPE (NEW AND EXISTING) SHALL BE PRESSURE TESTED PRIOR TO BUILDING OCCUPANCY.
5. EXISTING GAS PIPING SIZE VERIFIED USING TABLE 402.4(5) OF THE INTERNATIONAL FUEL GAS CODE USING THE LONGEST LENGTH METHOD 402.4.1 USING THE FOLLOWING INFORMATION:
   - INLET PRESSURE OF 2.0 PSI
   - PRESSURE DROP OF 1.0 IN. W.C. FOR SCHEDULE 40 METALLIC PIPE
   - MAXIMUM DEVELOPED LENGTH OF 100 FT.

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**FIFTH THIRD BANK**

Project Number: 26032
Project Name:
**FIFTH THIRD BANK WATSON**
**1,900 SF**

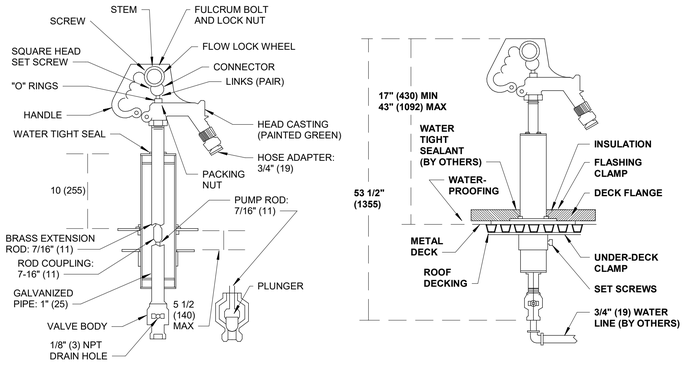
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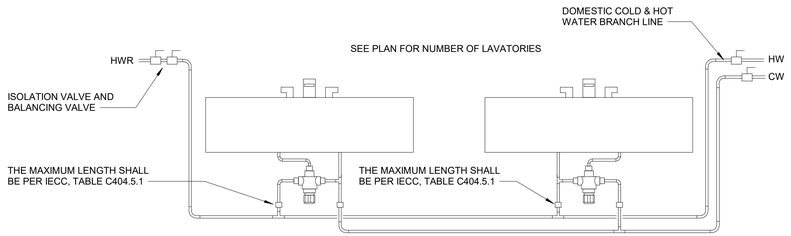
Sheet Title:
**PLUMBING NATURAL GAS ISO PLAN**

Sheet Number:
**P-403**

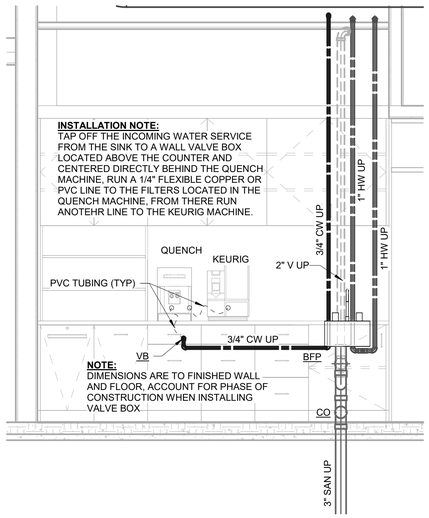
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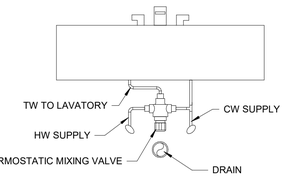

1 P-501 DETAIL OF ROOF HYDRANT
NTS



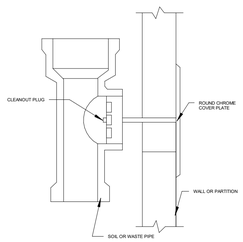

2 P-501 PUBLIC LAVATORY SUPPLY DETAIL
NOT TO SCALE



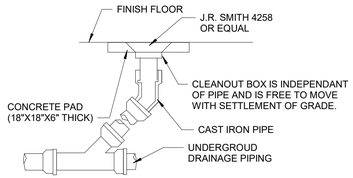

3 P-501 LIBRARY COUNTERTOP DRINKING EQUIPMENT ELEVATION
1/2" = 1'-0"



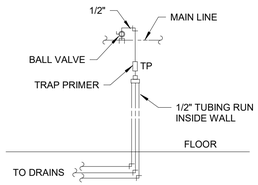

4 P-501 DETAIL OF TMV BELOW LAVATORY
NOT TO SCALE



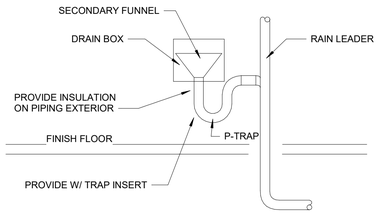

5 P-501 WALL CLEANOUT DETAIL
12" = 1'-0"



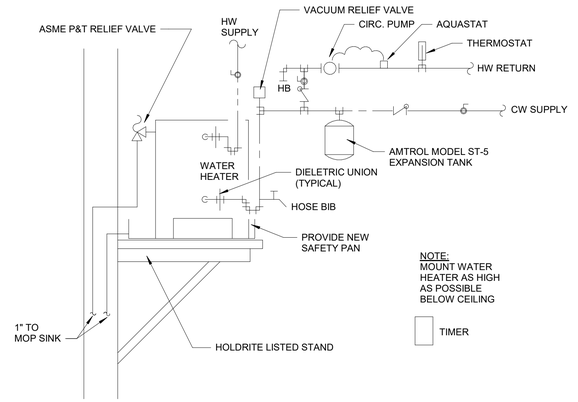

6 P-501 DETAIL OF CLEANOUT TO GRADE
NOT TO SCALE



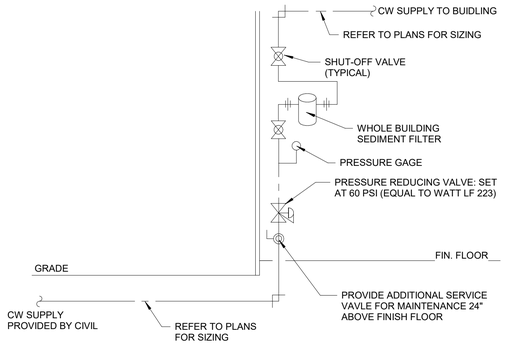

7 P-501 DETAIL OF PIPING AT TRAP PRIMER
NOT TO SCALE



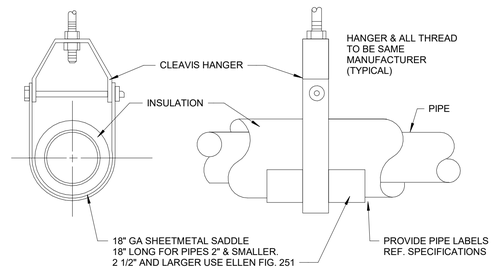

8 P-501 CONDENSATE DRAIN TO RAIN LEADER DETAIL
NOT TO SCALE




9 P-501 DETAIL OF PIPING AT WATER HEATER
NOT TO SCALE




10 P-501 DETAIL FOR WATER ENTRY
12" = 1'-0"




11 P-501 SUSPENDED PIPE SUPPORT
NOT TO SCALE

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Project Number: 26032
Project Name:
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**1,900 SF**

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Key Plan:

Sheet Title:
**PLUMBING DETAILS**

Sheet Number:
**P-501**